1) grain growth

- what is the UO_2 grain size after 4 hrs @ 1500 K?

$D_0 = 3\ \mu m$ $\quad M_0 = 4.6\times10^{-9}\ \frac{m^4}{J\cdot s}$ $\quad Q = 2.77\ eV$ $\quad \gamma_{GB} = 1.6\ J/m^2$

$$D(t)^2 - D_0^2 = 2 M_{GB}\, \gamma_{GB}\, t$$

$$M_{GB} = M_0 \exp\left(-Q/kT\right) = 4.6\times10^{-9} \exp\left(\frac{-2.77}{8.617\times10^{-5} \times 1500}\right)$$

$$= 2.27\times10^{-17}\ \frac{m^4}{J\cdot s}$$

$$D(t)^2 - \left(8\times10^{-6}\right)^2 = 2\left(2.27\times10^{-17}\right)(1.6)\left(4\times60\times60\right)$$

$$D(t)^2 = 1.046\times10^{-11} + 6.4\times10^{-11}$$

$$\boxed{D = 8.63\ \mu m}$$

2) Densification

- how much densification has occurred after 5 days?

- $\dot{F} = 8\times10^{12}\ \frac{fiss}{cm^3\cdot s}$ $\quad \rho(UO_2) = 10.97\ g/cc$ $\quad T = 1200\ K$ $\quad \Delta\rho_0 = 0.01$

$\beta_D = 5\ MWD/kgU \to 0.0053$

$$\epsilon_D = \Delta\rho_0 \left(\exp\left(\frac{\beta \ln 0.01}{C_D \beta_D}\right) - 1\right)$$

- what is current burnup? $\quad \beta = \frac{\dot{F} t}{N_u}$ $\quad N_u = \frac{10.97\ g/cc}{270\ g} \cdot [illegible]\times10^{23}\ \frac{1\ U}{1\ UO_2}$

$N_u = 2.45\times10^{22}\ [illegible]$

$$\beta = \frac{\left(5\times10^{12}\right)\left(5\times24\times60\times60\right)}{2.45\times10^{22}} = 8.8\times10^{-5}\ \text{FIMA}$$

$$\epsilon_D = 0.01\left(\exp\left(\frac{8.8\times10^{-5}\ \ln 0.01}{1 \times 0.0053}\right) - 1\right) = \boxed{-0.00074}$$

- very minimal, but started

2) cont...

when long densification stop?

$$\delta = \delta_0 = \frac{\dot{F} t}{N_4} \Rightarrow 0.0057 = \frac{(5\times10^{12})\, t}{245\times10^{20}}$$

$t = 2.597\times10^{7}\,s$

$\underline{t = 300 \text{ dys}}$

---

5) Thin-walled stresses $\bar{R}_c = 0.6$ cm $p = 20$ MPa $t_c = 0.05$ cm

$$\sigma_\theta = \frac{pR}{\delta} = \frac{20(0.6)}{0.05} = 240 \text{ MPa}$$

$$\sigma_z = \frac{pR}{2\delta} = \frac{20(0.6)}{2(0.05)} = 120 \text{ MPa}$$

$$\sigma_r = -\frac{p}{2} = \frac{-20}{2} = -10$$

4) Thick-walled, same problem $r = 0.58$ cm

$\bar{R}_c = 0.6$ cm

$t_c = 0.05$ cm

$$\sigma_\theta = p\left[\frac{(R_o/r)^2+1}{(R_o/R_i)^2-1}\right] = 20\left[\frac{\left(\frac{0.625}{0.58}\right)^2+1}{1.087^2-1}\right] = \underline{238 \text{ MPa}}$$

$R_i = 0.575$ cm

$R_o = 0.625$ cm

$R_o/R_i = 1.087$

$$\sigma_z = \frac{p}{(R_o/R_i)^2-1} = \frac{20}{(1.087)^2-1} = \underline{110.1 \text{ MPa}}$$

$$\sigma_r = -p\left[\frac{(R_o/r)^2-1}{(R_o/R_i)^2-1}\right] = -20\left[\frac{\left(\frac{0.625}{0.58}\right)^2-1}{1.087^2-1}\right] \quad \underline{-17.7 \text{ MPa}}$$

5) Thickness of wall where $\sigma_\theta > \sigma_{fr}$

$\sigma_{fr} = 150$ MPa → thin wall, can do thick also...

$$\sigma_\theta = \sigma_f = \frac{pR}{\delta} \Rightarrow 150 = \frac{20(0.6)}{\delta}$$

$\underline{\delta = 0.008 \text{ cm}}$

b) Thermal stresses in cladding

LHR: 250 W/cm $\quad R_f: 0.45\,cm \quad R_o: 0.55\,cm \quad R_i: 0.5\,cm$

$\alpha: 8\times10^{-6}\ 1/K \quad E: 80\ GPa \quad \nu: 0.4 \quad @\ r: R_o \quad K_c: 0.15\ W/cm\,K$

$$\sigma_\theta: \frac{\Delta T}{2}\frac{\alpha E}{1-\nu}\left(\frac{r}{R_i}-1\right)\left(1-\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$$\Delta T = T_{co}-T_{ci} = \frac{LHR}{2\pi R_c}\frac{t_c}{K_c} = \frac{250}{2\pi\cdot 0.45}\frac{0.05}{0.15} = 29.5\ K$$

$$\sigma_\phi: \frac{29.5}{2}\frac{(8\times10^{-6})(80)}{1-0.4}\left(\frac{0.55}{0.5}-1\right)\left(1-\frac{0.5}{0.05}\left(\frac{0.55}{0.5}-1\right)\right)$$

$$\sigma_\phi: 0.0157\,(0.1)\,(0) = 0$$

$$\sigma_\theta: \frac{\Delta T}{2}\frac{\alpha E}{1-\nu}\left(1-2\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$$= 0.0157\left(1-2\frac{0.5}{0.05}\left(\frac{0.55}{0.5}-1\right)\right) = -0.0157\ GPa$$

$$\boxed{= -15.7\ MPa}$$

7) Thermal stress in fuel

$LHR = 250 \ W/cm$ $\quad R_f = 0.45 \ cm$ $\quad R_o = 0.55 \ cm$ $\quad R_i = 0.5 \ cm$

$\alpha_f = 11 \times 10^{-6} \ 1/K$ $\quad E = 200 \ GPa$ $\quad \nu = 0.35$ $\quad$ @ $r = 0.2$ $\quad K_f = 0.03 \ W/cm\text{-}K$

$$\sigma_\theta = -\sigma^* (1 - 3\eta^2) \qquad \sigma^* = \frac{\alpha E \Delta T}{4(1-\nu)}$$

$$\sigma_r = -\sigma^* (1 - \eta^2)$$

$$\Delta T = T_0 - T_s = \frac{LHR}{4\pi K_f} = \frac{250}{4\pi \cdot 0.03} = 663 \ K$$

$$\sigma_z = -2\sigma^* (1 - 2\eta^2)$$

$$\sigma^* = \frac{(11 \times 10^{-6})(200)(663)}{4(1 - 0.35)} = 786 \ MPa$$

$$\eta = \frac{r}{R_f} \Rightarrow \frac{0.2}{0.45} = 0.444$$

$$\sigma_\theta = -786 \left(1 - 3(0.444)^2\right) = \underline{-[illegible] \ MPa}$$

$$\sigma_r = -786 \left(1 - 0.444^2\right) = \underline{-631 \ MPa}$$

$$\sigma_z = -2(786)\left(1 - 2(0.444)^2\right) = \underline{-952 \ MPa}$$

---

8) $\Delta \delta_g = \Delta \bar{R}_c - \Delta R_f = \alpha_c \bar{R}_c \Delta T_c - \alpha_f R_f \Delta T_f$

$T_{ref} = 300K$ .... $\qquad LHR = 250 \ W/cm$